Grant P. Holzemer

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Experience

Google DeepMind | Machine Learning & Experimentation Platform | *March 2026 - Present*
Software Engineer III

- Proposed and led a re-architecture of XManager, the platform researchers use to launch and track ML experiments. Split 5 critical microservices into compute-isolated zones aligned with researcher workflows, so a failure in one workload no longer disrupts other researchers' experiments. Two components complete. Three in progress.
- Identified that the experimentation, compute, and ML serving platforms had no shared definition of a "hero run" (a critical, long-running training job). Proposed and now lead a cross-team initiative that sends a workload-criticality signal from XManager to the compute and serving platforms, so researchers' most important runs get prioritized compute and more precise alerting.
- Recognized that researchers had no client- or server-side monitoring for their high-priority training runs. Designed and built a self-serve Python alerting framework covering quota exhaustion, downstream failures, and client errors, now used by 15 research teams across Google DeepMind.
- Defined SLOs and escalation policies for the experiment analytics platform, which researchers use to analyze and compare their experiments. Drove its production readiness review, enabling handoff to SRE on-call.

Amazon Web Services | EC2 VPC | *June 2023 - March 2026*
Software Development Engineer II | *September 2025 - March 2026*
Software Development Engineer I | *January 2024 - August 2025*
Software Development Engineer Intern | *June 2023 - August 2023*

- Led design and implementation of cross-VPC Security Group referencing, a high-risk EC2 control-plane change spanning multiple mutating and read-only APIs, resolving edge cases in authorization, dependency ordering, and backwards compatibility.
- Owned customer-facing fixes to EC2 Security Group control-plane APIs, introducing batched updates and resilient fallbacks that eliminated customer production network outages during rule updates.
- Prevented breaking changes for 500+ customer accounts by identifying affected usage ahead of new networking launches and gating rollout with targeted controls.
- Reduced large-scale consistency validation runtime by ~90% by redesigning data access patterns with time-based indexing in DynamoDB, cutting regional data processing from months to weeks.
- Cut recovery time for a full regional outage from days to ~1 hour by automating large-scale state reprocessing.
- Built automated multi-region build and deployment pipelines that eliminated manual operator steps.
- Served as primary on-call for EC2 Security Groups and other control-plane services, resolving 400+ tickets including high-severity incidents; authored and led post-incident reviews that drove anomaly detection, rollback tooling, and better observability.
- Mentored two summer interns and multiple new hires while authoring onboarding documentation, architectural overviews, and operational runbooks to improve ramp-up time and long-term team health.

Raytheon Intelligence & Space | **Software Engineer Intern** | May 2022 - August 2022

- Engineered a cross-platform Electron application to track ~3,000 active and decayed commercial satellites, using Python, Angular, and MySQL, hosted on AWS.
- Integrated server-side STK processing to regularly update satellite models and data for real-time tracking.

Skills

Languages: Java, C++, Python, Ruby, TypeScript, JavaScript, SQL
Infrastructure & Cloud: AWS (EC2, ECS, VPC, DynamoDB, Lambda, S3, CloudFormation, CloudWatch)
Systems: Distributed Systems, Networking (gRPC, HTTP, TCP/UDP/ICMP, RDMA), API Design, Reliability Engineering
Tools: Git, CI/CD Pipelines, IaC, Cloud Monitoring & Alerting, Grafana

Education

West Virginia University - B.S. Computer Science, Magna Cum Laude (GPA 3.78/4.0) | May 2023

Military Service and Awards

- U.S. Army National Guard, Specialist (E4), Power-Generation Equipment Repairer | June 2020 - June 2026
- Certificate of Recognition from West Virginia Governor Jim Justice
- WVU Foundation Outstanding Senior (2023), one of 52 graduating seniors university-wide recognized for scholarship, leadership, and service